

Fourier measurements are optimal for CGLMP Bell tests on maximally entangled states in every dimension

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Abstract

Bell inequalities certify that correlations between distant measurements have no local explanation, and in high dimensions they underpin device-independent tests of entanglement and randomness. The Collins–Gisin–Linden–Massar–Popescu (CGLMP) inequalities are the standard Bell inequalities for two parties with d outcomes per measurement. On the maximally entangled state of two d -level systems the largest known violation is produced by Fourier-type measurements found by Durt, Kaszlikowski and Żukowski (DKZ). Whether other measurements can do better was posed as part of Open Quantum Problem 27 of IQOQI Vienna. Here we prove that, for every number of outcomes d and for maximally entangled states of any local dimension, no projective measurements exceed the DKZ value, and that the DKZ measurements are the only optimal ones up to local unitaries and an inert auxiliary system. The key new ingredient is a matrix inequality on a strip in the complex plane, which we prove for matrices of every size by exhibiting an explicit positive density in a two-variable extension of Stahl’s theorem, formerly the Bessis–Moussa–Villani conjecture. A computer-assisted certificate, checked in interval arithmetic, connects this inequality to the Bell problem for every d . For $d \leq 20$ the entire proof, including this certificate, is checked by the Lean theorem prover. We also give partial answers to the two other clauses of Part B, on noise resistance and on Kullback–Leibler discrimination.

Bell’s theorem states that the correlations between measurements on distant quantum systems can be stronger than any local hidden-variable model allows^{1–3}. Loophole-free experiments have confirmed such violations^{4–6}, and a violation can certify, without trusting the devices, that entanglement is present, that outcomes are random or that a key is secret^{7,8}. Each of these applications rests on a quantitative fact: for a given Bell expression one must know the largest value that quantum mechanics allows, and which states and measurements attain it. For two parties with two settings and two outcomes this is Tsirelson’s bound⁹ for the Clauser–Horne–Shimony–Holt inequality².

High-dimensional entanglement is now produced routinely in photonic platforms^{10,11}, and Bell tests with many outcomes can tolerate more noise and lower detection efficiencies than tests with qubits^{12–14}. The canonical Bell inequalities for this setting were found by Collins, Gisin, Linden, Massar and Popescu (CGLMP)¹³, with an equivalent form for $d = 3$ by Kaszlikowski and co-workers¹⁵. For the maximally entangled state of two d -level systems, the measurements found numerically by Kaszlikowski *et al.*¹² and described analytically by Durt, Kaszlikowski and Żukowski (DKZ)¹⁶, namely phase shifts followed by a discrete Fourier transform, give the largest known CGLMP value. Numerical searches never found anything better^{12,13,16}. Whether the DKZ measurements are necessarily the optimal ones was asked by Gill¹⁷ and is Part B of Open Quantum Problem 27 of IQOQI Vienna¹⁸, a list of problems that goes back to ref. 19.

Here we answer the maximally entangled part of that question for every number of outcomes d . No projective measurements on a maximally entangled state of any local dimension exceed the DKZ value, and

the DKZ measurements are the only optimal ones up to local unitaries and an inert auxiliary system. The proof combines an exact reduction to a long-range clock model, a new inequality for non-normal matrices whose real part is a projection, and a computer-assisted certificate for a discrete step. The matrix inequality is the heart of the argument. We prove it for matrices of every size by uncovering an explicit positive density behind a two-variable extension of the Laplace representation of Bessis, Moussa and Villani (BMV), which was conjectured in 1975 and proved by Stahl^{20,21}. The argument has also been formalised in the Lean theorem prover^{22,23}. The formal proof is complete except for one input, the certificate for $d \geq 21$. For $d \leq 20$ Lean checks the certificate as well, so for these d the result is machine-checked with no hypotheses. The matrix inequality and the positivity theorem behind it are formalised in full, for matrices of every size. We also give partial answers to the two other clauses of Part B, on noise resistance and on Kullback–Leibler discrimination (Table 1). These results have not yet been reviewed by outside experts.

The CGLMP inequality and the question

Alice and Bob each choose one of two settings, $x, y \in \{1, 2\}$, and obtain outcomes $A_x, B_y \in \mathbb{Z}_d = \{0, \dots, d-1\}$ (Fig. 1a). The CGLMP expression is¹³

$$I_d = \sum_{k=0}^{\lfloor d/2 \rfloor - 1} \left(1 - \frac{2k}{d-1}\right) \left([P(A_1 = B_1 + k) + P(B_1 = A_2 + k + 1) + P(A_2 = B_2 + k) + P(B_2 = A_1 + k)] \right. \\ \left. - [P(A_1 = B_1 - k - 1) + P(B_1 = A_2 - k) + P(A_2 = B_2 - k - 1) + P(B_2 = A_1 - k - 1)] \right), \quad (1)$$

where $P(A_x = B_y + k)$ is the probability that $A_x - B_y \equiv k \pmod{d}$. The four pairs of settings form a chain A_1, B_1, A_2, B_2, A_1 . The expression rewards outcomes that are close on the d -hour clock in one direction along the chain and penalizes the other direction, and the extra shift by one in the term $P(B_1 = A_2 + k + 1)$ makes it impossible for a local model to be perfectly correlated all around the chain. Local hidden-variable models obey $I_d \leq 2$ ¹³. For $d = 2$ the expression reduces to the Clauser–Horne–Shimony–Holt expression.

We consider the maximally entangled state $|\Phi_D\rangle = D^{-1/2} \sum_{j=1}^D |j\rangle|j\rangle$ of two systems of an arbitrary dimension D , and projective measurements $\{P_a^x\}_{a \in \mathbb{Z}_d}$ for Alice and $\{Q_b^y\}_{b \in \mathbb{Z}_d}$ for Bob, so that $P(A_x = a, B_y = b) = \langle \Phi_D | P_a^x \otimes Q_b^y | \Phi_D \rangle$. The DKZ measurements act on $D = d$: each party applies the phases $e^{2\pi i j \theta / d}$ to the basis states $|j\rangle$, with $\theta = 0, \frac{1}{2}$ for Alice’s two settings and $\theta = \frac{1}{4}, -\frac{1}{4}$ for Bob’s, then a discrete Fourier transform, and measures in the computational basis. They give^{13,16}

$$I_{\text{ME}}(d) = \frac{4}{d(d-1)} \sum_{j=1}^{d-1} (d-j) \sec \frac{\pi j}{2d}, \quad (2)$$

for example $I_{\text{ME}}(2) = 2\sqrt{2}$, $I_{\text{ME}}(3) = 2.8729\dots$ and $I_{\text{ME}}(10) = 2.9398\dots$. As $d \rightarrow \infty$, $I_{\text{ME}}(d)$ tends to $32G/\pi^2 = 2.9698\dots$, where G is Catalan’s constant (Fig. 1c).

Two features make this problem natural and hard. First, it is a statement about the maximally entangled state, the natural target state of high-dimensional sources and the state for which the DKZ measurements were designed. It is not the same as the maximal CGLMP value over all states: for $d \geq 3$ that maximum is attained by states that are not maximally entangled^{24–26}, and it has been determined exactly only for small d (refs 27,28). Second, the measurement operators of the two settings do not commute and may act on a space of any dimension. A numerical search can only explore finitely many dimensions, and the semidefinite

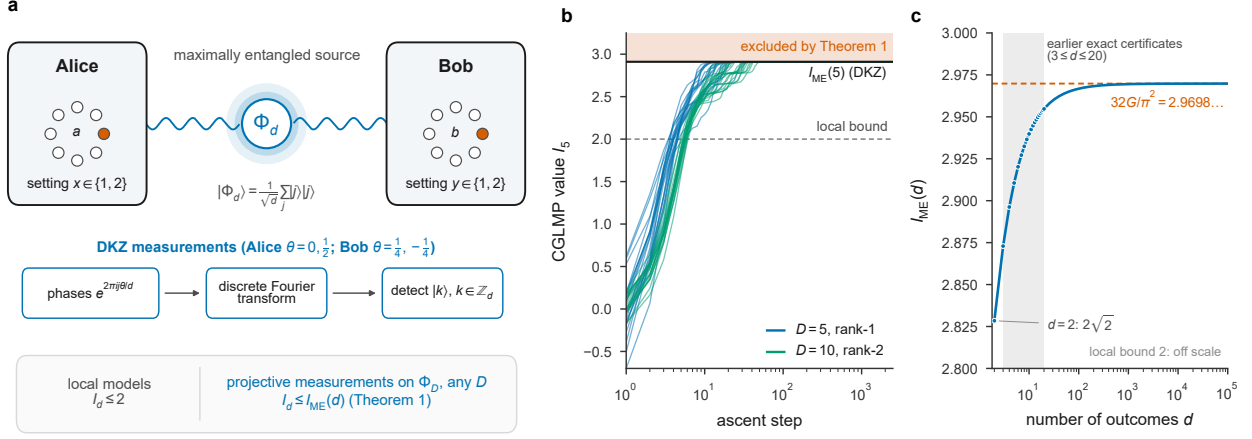


Figure 1. The CGLMP Bell test on maximally entangled states. **a**, A source distributes the maximally entangled state Φ_d . Alice and Bob each choose one of two settings and record one of d outcomes; the CGLMP value I_d (equation (1)) rewards outcomes that are close on the d -hour clock. The DKZ measurements apply setting-dependent phases, a discrete Fourier transform and a computational-basis measurement. Local models obey $I_d \leq 2$; Theorem 1 shows that every projective measurement on a maximally entangled state of any dimension obeys $I_d \leq I_{ME}(d)$. **b**, Gradient ascent of I_5 over projective strategies on Φ_D from 40 random starting points, 20 for each D ($D = 5$, rank-one projectors, blue; $D = 10$, rank-two projectors, green). All runs stop at the DKZ value $I_{ME}(5) = 2.9105 \dots$ (black line); the largest final value exceeds it by 4×10^{-13} , the floating-point accuracy. The shaded region above $I_{ME}(5)$ is excluded by Theorem 1. **c**, $I_{ME}(d)$ for $2 \leq d \leq 10^5$. It approaches $32G/\pi^2$ (dashed) from below, as $32G/\pi^2 - 0.3034/d$ for large d ; dots mark $d \leq 20$, and the grey band the range $3 \leq d \leq 20$ covered by our earlier exact certificates²⁸.

hierarchies that bound quantum correlations²⁹ certify one value of d at a time.

Before this work the evidence was as follows. For $d = 2$ the statement is Tsirelson's bound. For $d = 3$ a semidefinite relaxation agrees with the DKZ value to within 10^{-10} (ref. 30); this is the partial result listed in ref. 18. In earlier work we obtained exact sum-of-squares certificates, with cyclotomic coefficients, proving optimality and rigidity of DKZ among projective measurements for $3 \leq d \leq 20$ ²⁸, and we proved the statement for every d in the special case in which the four link operators of the strategy coincide, which makes the problem commutative³¹. Within a family of Fourier measurements with linear phase offsets, the DKZ point was recently found numerically to be a strict local maximum for $d \leq 6$ ³². The certificates grow quickly with d and show no pattern that could be extrapolated, and relaxations that only use correlations among a bounded number of settings fail for large d , even in the commutative case³¹.

Optimality and rigidity in every dimension

Our main results are the following.

Theorem 1 (optimality). *For every $d \geq 2$, every $D \geq 1$ and all projective measurements of Alice and Bob on $\mathbb{C}^D \otimes \mathbb{C}^D$, the CGLMP value of the maximally entangled state Φ_D satisfies $I_d \leq I_{ME}(d)$. The DKZ measurements attain this value.*

Theorem 2 (rigidity). *If projective measurements on Φ_D attain $I_d = I_{ME}(d)$, then d divides D , and there is a unitary $u : \mathbb{C}^D \rightarrow \mathbb{C}^d \otimes \mathbb{C}^{D/d}$ such that the local unitary $u \otimes \bar{u}$ maps the measurements to the DKZ measurements tensored with the identity on $\mathbb{C}^{D/d}$, and maps Φ_D to $\Phi_d \otimes \Phi_{D/d}$. The unitary u is unique up to $u \mapsto (\mathbf{1}_d \otimes u')u$. In particular, if d does not divide D , the maximal CGLMP value on Φ_D is strictly smaller than $I_{ME}(d)$.*

Theorem 1 identifies the largest CGLMP value of maximally entangled states for every d : it is the Tsirelson-type bound of the CGLMP expression restricted to these states. Theorem 2 says that this value is attained in an essentially unique way. Observing $I_d = I_{\text{ME}}(d)$ with projective measurements on a maximally entangled state certifies the measurements up to a local change of basis and an inert auxiliary system. This is a self-testing statement^{33,34} for measurements on maximally entangled states. It is weaker than full device-independent self-testing, which would also have to certify the state. Device-independent self-tests of maximally entangled states of every dimension are known for other Bell expressions, which are tailored to these states^{35–37}; they do not determine the maximal value of I_d . Theorem 1 also gives a dimension-free test. Within projective measurements, a value $I_d > I_{\text{ME}}(d)$ rules out a maximally entangled source of any local dimension.

Figure 1b illustrates the content of Theorem 1 for $d = 5$. Gradient ascent over strategies started from random measurements climbs from values near zero, crosses the local bound and stops at the DKZ value in every run, for rank-one and rank-two projectors alike. Such numerics explore only finitely many dimensions and starting points, and they find local maxima at best. Theorems 1 and 2 hold for every d and every D .

The proofs of Theorems 3 and 4 below, which carry the new mathematics, are analytic. The proofs of Theorems 1 and 2 in addition use a rigorous computer-assisted certificate for a family of inequalities between explicit kernels (CONE_d , below). It is verified in interval arithmetic for every d : by explicit certificates for $d \leq 2000$ and by a proof with certified error bounds for $d \geq 2001$. All other steps are also verified in Lean, and so is CONE_d for $d \leq 20$ (Methods). Theorems 1 and 2 are therefore proved for every d , with a computer-assisted step in interval arithmetic for $d \geq 21$, and fully formalised for $d \leq 20$; Theorems 3 and 4 are formalised in full. The statements concern projective measurements; general positive-operator-valued measurements are not covered (Methods).

From Bell strategies to a clock model

The first step is an exact reformulation³¹. Write $w = e^{2\pi i/d}$ and encode Alice’s measurements by the unitaries $U_x = \sum_a w^a P_a^x$ and Bob’s by $U'_y = \sum_b w^b Q_b^y$. The CGLMP expression is invariant under a cyclic group of order $4d$ that shifts the outcomes and rotates the chain of settings. Averaging over this group and applying the Stone–von Neumann theorem for $\mathbb{Z}_d$ (Methods), every strategy on Φ_D becomes a family of d unitaries $V_0, \dots, V_{d-1}$ on an auxiliary space, each of order four ($V_k^4 = \mathbf{1}$), with

$$I_d = \frac{4F(V)}{d(d-1)}, \quad F(V) = \sum_{0 \leq j < k \leq d-1} \text{Re}[h_{k-j} \text{tr}(V_j V_k^\dagger)], \quad h_m = \sec \frac{\pi m}{2d} - i \csc \frac{\pi m}{2d}, \quad (3)$$

where tr is the normalized trace. Conversely every such family comes from a strategy. The DKZ strategy corresponds to $V_0 = \dots = V_{d-1} = \mathbf{1}$, so Theorem 1 is equivalent to $F(V) \leq F_{\text{DKZ}} := \sum_{m=1}^{d-1} (d-m) \sec \frac{\pi m}{2d}$.

Equation (3) describes a long-range clock model: d sites carry four-state “spins” V_k , and every pair of sites is coupled, with a complex coupling h_{k-j} that depends on their distance. If the V_k commute they can be diagonalized together, and the model becomes a classical spin model. Its ground state is a single domain wall, which we showed for every d in ref. 31. The quantum problem asks whether matrix-valued spins can do better than classical ones. They cannot, but the argument has to deal with genuinely non-commuting configurations: numerically, the reduced model has non-commuting local maxima that are not global maxima, and the classical proof does not transfer.

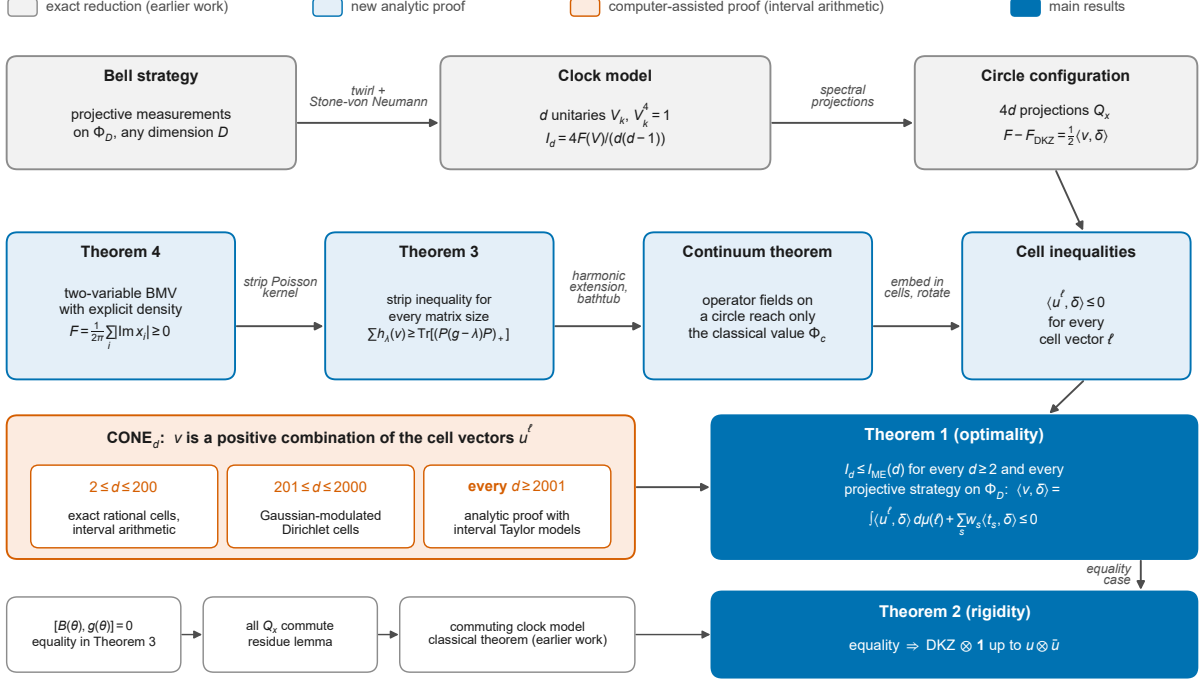


Figure 2. Architecture of the proof. Top row: exact reduction of a projective strategy on Φ_D to the clock model (3) and to a configuration of $4d$ projections on a circle, in which the deficit $F - F_{\text{DKZ}}$ becomes the linear form $\frac{1}{2} \langle v, \delta \rangle$ of equation (12) (Methods). Middle row: the new analytic chain. Theorem 4 (an explicit positive density) implies Theorem 3 (the strip inequality, for matrices of every size); Theorem 3 gives the continuum theorem for operator fields on a circle, and embedding the discrete configuration with cells of lengths ℓ gives one linear inequality $\langle u^\ell, \delta \rangle \leq 0$ for every cell vector ℓ . Bottom: CONE_d , a statement about explicit Clausen-function kernels only, is certified for every d in three regimes. Together they give Theorem 1. Following the equality case through the chain gives Theorem 2.

A second reformulation explains where the difficulty sits. Collect the spectral projections of all V_k into $4d$ projections Q_x placed at the points x of a circle with $4d$ sites (Methods). The function F then becomes a pairing of the projections at x with those at $x + d$ through the kernel $\cot(\pi(x - y)/4d)$, which is the kernel of the discrete Hilbert transform. In a continuum limit, in which the circle becomes continuous and the projections become operator-valued functions, the classical theory is a theorem of Stein and Weiss: the distribution of the conjugate function of an indicator function depends only on the measure of the set³⁸. This theorem determines the continuum optimum, and the discrete problem differs from it by terms at the junctions between blocks of the configuration. For commuting spins these two parts can be handled separately³¹. For matrices, neither the continuum statement nor the discrete correction was available. Figure 2 summarizes how we obtain both.

A matrix inequality on a strip

The continuum theorem needs a single inequality between matrices. Let B be an orthogonal projection on $\mathbb{C}^M$, let $P = 1 - B$, and let g be Hermitian. The matrix $X = B + ig$ is not normal in general, but its eigenvalues lie in the strip $0 \leq \text{Re } v \leq 1$, because $\text{Re} \langle \psi, X\psi \rangle = \langle \psi, B\psi \rangle$. On this strip let h_λ be the harmonic function with boundary values $(y - \lambda)_+ = \max(y - \lambda, 0)$ on the left edge and 0 on the right edge (Fig. 3b). It has the closed form

$$h_\lambda(x + iy) = -\frac{1}{\pi^2} \text{Im Li}_2(e^{\pi(y-\lambda)} e^{-i\pi x}), \quad (4)$$

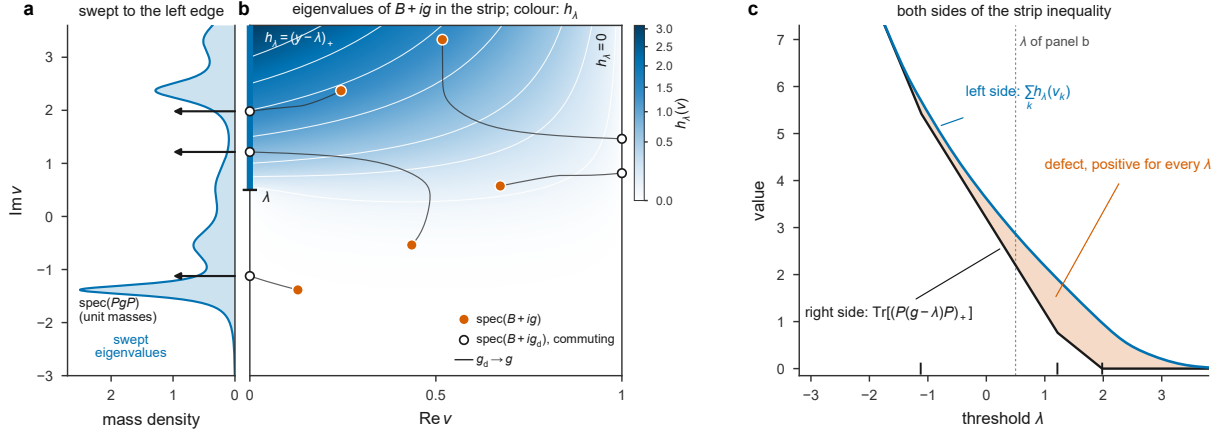


Figure 3. The strip inequality. A random example with $M = 5$, rank $B = 2$ (seed fixed in the figure script). **b**, The harmonic function h_λ of equation (4) on the strip $0 \leq \text{Re } \nu \leq 1$ for $\lambda = 0.5$ (colour; white contours), with boundary values $(y - \lambda)_+$ on the left edge (thick line) and 0 on the right edge. Orange dots: the eigenvalues of $B + ig$. Open circles: the eigenvalues for the pinched matrix $g_d = BgB + PgP$, which commutes with B ; they lie on the edges, at the spectra of PgP (left) and BgB (right). Thin lines: the eigenvalues of $B + i(g_d + t(g - g_d))$ for $0 \leq t \leq 1$. **a**, The eigenvalues of $B + ig$ swept to the left edge by the harmonic measure of the strip (blue density, total mass rank $P = 3$), compared with the three eigenvalues of PgP (black spikes). Both have the same mass and the same mean. Theorem 3 says that the blue distribution is more spread out, in the convex order. **c**, Both sides of inequality (5) as functions of λ . The right side (black) is piecewise linear with kinks at the eigenvalues of PgP (ticks). The defect (shaded) is positive for every λ ; for the commuting pinched matrix the two sides coincide.

1 where Li_2 is the dilogarithm.

2 **Theorem 3 (strip inequality).** For every M , every orthogonal projection B on $\mathbb{C}^M$, every Hermitian g and
3 every real λ ,

$$\sum_{\nu \in \text{spec}(B+ig)} h_\lambda(\nu) \geq \text{Tr}[(P(g - \lambda)P)_+], \quad (5)$$

4 where eigenvalues are counted with algebraic multiplicity and $(\cdot)_+$ is the positive part. Equality holds for
5 one, equivalently for all, λ if and only if B and g commute.

6 If B and g commute, the eigenvalues of $B + ig$ lie on the edges of the strip: $i\mu$ for every eigenvalue μ of
7 PgP on the range of P , and $1 + i\mu'$ for every eigenvalue μ' of BgB on the range of B . The boundary values
8 of h_λ then turn the left side of (5) into the right side exactly. If B and g do not commute, the eigenvalues
9 move into the interior (Fig. 3b), and the theorem says that this can only increase the left side. There is an
10 equivalent formulation that does not refer to λ . Sweep every eigenvalue ν to the left edge with the harmonic
11 measure of the strip, that is, replace it by the distribution of the point at which a Brownian motion started at
12 ν leaves the strip, counting only exits through the left edge (an eigenvalue with real part x contributes mass
13 $1 - x$). Theorem 3 states that the resulting distribution dominates the spectrum of the compression PgP in
14 convex order³⁹ (Fig. 3a,c). It is thus a majorization theorem of Schur–Horn type for non-normal matrices,
15 with the Stein–Weiss theorem as its commutative case.

16 Before finding the general proof, we could establish Theorem 3 only in special cases, for instance for
17 $M \leq 4$, by a geometric argument based on Pedoe’s inequality for polygons and independently by a Bellman-
18 function argument. The general case is the step from commuting to non-commuting measurements. Inequal-
19 ities of this kind are delicate. Several natural strengthenings, such as monotonicity of the left side along the
20 path from the pinched matrix $BgB + PgP$ to g , are false, with counterexamples already for $M = 3$.

A hidden positive density

Theorem 3 follows from an identity that exhibits its defect as an integral of a positive function. For complex a and t consider the trace function

$$\mathcal{D}(a, t) = \text{Tr} e^{ag-tP} - e^{-t} \text{Tr}_P e^{aPgP} - \text{Tr}_B e^{aBgB}, \quad (6)$$

in which the last two traces are over the ranges of P and B . The first term is the trace of an exponential of two non-commuting matrices; the other two are what it would be if g were replaced by its pinching $BgB + PgP$. For fixed $s \in \mathbb{R}$ and $0 < \tau < 1$ consider the matrix pencil $g - s - x(P - \tau)$ and its M roots $x_1(s, \tau), \dots, x_M(s, \tau)$, the values of x at which it is singular.

Theorem 4 (two-variable BMV formula with an explicit density). *Let*

$$F(s, \tau) = \frac{1}{2\pi} \sum_{i=1}^M |\text{Im} x_i(s, \tau)| \quad (s \in \mathbb{R}, 0 < \tau < 1). \quad (7)$$

Then F is continuous, non-negative and supported in $[\lambda_{\min}(g), \lambda_{\max}(g)] \times [0, 1]$, and for all complex a, t

$$\mathcal{D}(a, t) = a^2 \int_0^1 \int_{\mathbb{R}} e^{as-t\tau} F(s, \tau) ds d\tau. \quad (8)$$

Moreover, the defect of inequality (5) is

$$\sum_{\nu \in \text{spec}(B+ig)} h_\lambda(\nu) - \text{Tr}[(P(g - \lambda)P)_+] = \int_0^1 \int_{\mathbb{R}} K_{1-\tau}(s - \lambda) F(s, \tau) ds d\tau, \quad (9)$$

where $K_X(u) = \sin(\pi X)/(2 \cosh \pi u - 2 \cos \pi X)$ is the Poisson kernel of the strip.

For a projection P and $a = 1$, equation (8) contains Stahl's theorem^{21,40}, which settled the conjecture of Bessis, Moussa and Villani^{20,41}: the function $t \mapsto \text{Tr} e^{A-tP}$ is the Laplace transform of a positive measure. Here the measure is explicit. It consists of atoms $\text{Tr}_B e^{BAB}$ at $\tau = 0$ and $\text{Tr}_P e^{PAP}$ at $\tau = 1$, plus the absolutely continuous part with density $\int e^s F(s, \tau) ds$. The second variable a and the subtraction of the pinched terms are what the strip inequality needs, and positivity survives both. Theorem 4 is a statement about the positivity of a two-variable transform, so it also yields the strip inequality (5), which involves functions that are not exponentials. Stahl's theorem alone gives inequality (5) only for exponential test functions.

Why should such a density exist, and why is it positive? The roots x_i are the eigenvalues of $(P - \tau)^{-1}(g - s)$, a matrix that is self-adjoint for an indefinite inner product, so they are real or come in complex-conjugate pairs. For the pinched pencil all roots are real. The proof (Methods) reconstructs the distribution T whose Laplace transform is $\mathcal{D}/a^2$ from its one-dimensional projections, as in computed tomography⁴². These projections are the differences between the eigenvalue distributions of $g - \xi P$ and of the pinched matrix, and filtered back-projection turns them into the logarithm of the ratio of the two characteristic polynomials of the pencil. A Jensen-type identity shows that integrating this logarithm along a line removes all real roots and leaves exactly $\pi \sum_i |\text{Im} x_i|$. The density is therefore positive because non-commutativity can only create complex pairs of roots, each of which contributes with a positive sign. A Paley–Wiener argument⁴³ guarantees that T exists and lives on a rectangle, and the behaviour of $\mathcal{D}$ as $t \rightarrow \pm\infty$ excludes extra mass on

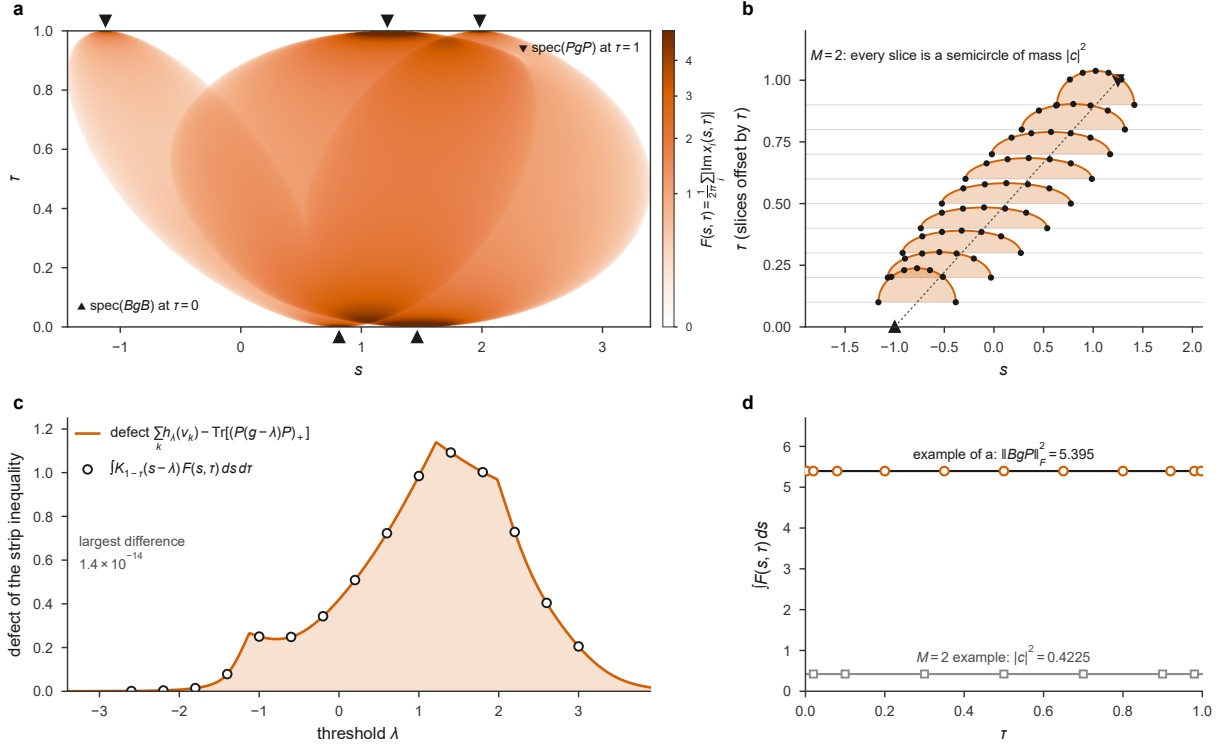


Figure 4. The explicit positive density of Theorem 4. **a**, $F(s, \tau)$ of equation (7) for the example of Fig. 3, computed from the roots of the pencil $g - s - x(P - \tau)$ (colour scale compressed by a square root). The density consists of lens-shaped sheets that connect the eigenvalues of BgB at $\tau = 0$ (triangles below) with those of PgP at $\tau = 1$ (triangles above). **b**, For $M = 2$, $B = \text{diag}(1, 0)$ and $g = \begin{pmatrix} \alpha & c \\ c & \beta \end{pmatrix}$, every slice is a semicircle law of radius $2|c|\sqrt{\tau(1-\tau)}$ centred at $\tau\beta + (1-\tau)\alpha$ (shaded curves, offset by τ ; here $\alpha = -1, \beta = 1.25, c = 0.65$); dots are the values computed from the roots. **c**, The defect of the strip inequality for the example of panel a, computed from the eigenvalues of $B + ig$ and PgP (line), and as the integral of the density against the strip Poisson kernel (circles, adaptive quadrature); the largest difference is 1.4×10^{-14} . **d**, Every horizontal slice of the density has the same mass, $\int F(s, \tau) ds = \|BgP\|_F^2$ (circles: example of panel a; squares: $M = 2$, where the mass is $|c|^2$).

1 the edges $\tau = 0, 1$, so $T = F$. A second, elementary proof (Methods) obtains the projections of F directly
2 from a Burgers equation obeyed by the roots, without distributions; it is the proof checked in Lean.

3 The same mechanism, a two-dimensional Fourier inversion of a trace of a pencil whose density is given
4 by the imaginary parts of pencil eigenvalues, was introduced by Heinävaara to give a new proof of Stahl's
5 theorem⁴⁴. Theorem 4 is a variant adapted to a projection and to the pinched subtraction. We found it
6 independently and learned of the connection afterwards. Figure 4 shows the density for the example of
7 Fig. 3, the semicircle laws that appear for $M = 2$, and two independent numerical checks. The defect equals
8 the strip transform of F to 1.4×10^{-14} , and every horizontal slice of F has mass $\|BgP\|_F^2$, a sum rule that
9 the proof does not use.

10 From the strip back to the Bell inequality

11 With Theorem 3 available for matrices of every size, the continuum problem can be solved. Let $B(\theta)$ be
12 a projection-valued function on the circle, let $0 \leq A(\theta) \leq 1 - B(\theta)$, and let g be the conjugate function
13 (periodic Hilbert transform) of B . The *continuum theorem* states that

$$\int \text{tr}(A(\theta)g(\theta)) \frac{d\theta}{2\pi} \leq \Phi_c(\alpha, \beta) = \frac{\text{Cl}_2(2\pi\alpha) + \text{Cl}_2(2\pi\beta) + \text{Cl}_2(2\pi(1 - \alpha - \beta))}{2\pi^2}, \quad (10)$$

where α and β are the mean normalized traces of A and B , and Cl_2 is the Clausen function. The right side is the value of two adjacent arcs, so in the continuum, matrix-valued configurations do no better than classical ones. The proof (Methods) extends $B + ig$ to an analytic matrix function on the disc whose eigenvalues stay in the strip, uses that the harmonic function h_λ of these eigenvalues has the correct mean, applies Theorem 3 at every point of the circle and finishes with the bathtub principle.

To return to the discrete problem, we embed the configuration of $4d$ projections into the continuous circle, giving the site x an arc whose length ℓ_x depends only on x modulo d . The constraints of the Bell problem make every such embedding admissible for the continuum theorem, and averaging over rotations turns inequality (10) into one linear inequality $\langle u^\ell, \delta \rangle \leq 0$ for the vector δ that measures the deviation of the configuration from DKZ. The vector u^ℓ depends only on the cell lengths and is given explicitly by Clausen functions (Methods). Equal cell lengths are not enough: the corresponding u^ℓ is not proportional to the vector v that defines $F - F_{\text{DKZ}}$, and the difference is the junction effect that defeats a direct continuum argument. Unequal cells repair this. Random cells, distributed like the gaps between uniform random points on the circle, reproduce the singular part of the discrete kernel exactly, through a Bernstein-polynomial identity (Methods), and leave a smooth residual of relative order $1/d$. What is needed, and what we prove, is

CONE_d: the vector v is a positive combination of vectors u^ℓ with all cells of positive length (and of trivially admissible directions).

Then $F - F_{\text{DKZ}} = \frac{1}{2} \langle v, \delta \rangle$ is a positive combination of quantities that the continuum theorem shows to be non-positive, and Theorem 1 follows. CONE_d involves only explicit special functions, with no operators and no quantum mechanics.

We certify CONE_d for every d in three regimes (Fig. 5). For $2 \leq d \leq 200$ we give explicit rational cell vectors (all cell lengths multiples of $1/16$) and positive weights, and verify the vector identity in interval arithmetic with rigorous enclosures of the Clausen function. For $201 \leq d \leq 2000$ we use random cells whose Dirichlet parameters are modulated by a stationary Gaussian process. The required identity then decouples into one scalar equation per distance, its solution is enclosed in interval arithmetic for each d , and a Poincaré–Miranda argument⁴⁵ passes from the decoupled equations to the exact ones. Each of these 1800 certificates is produced by a single run of one checker at one precision, and every margin is certified on the Poincaré–Miranda box itself. For $d \geq 2001$ the same construction is treated analytically. The solution is approximated by a smooth model, the fourth differences that control positivity are bounded by interval Taylor models on boxes that cover every $d \geq 2001$ at once, and the error of the model is bounded pointwise. All margins are large. The smallest certified weights for $d \leq 200$ exceed $0.08/d$, the fixed-point derivative bound exceeds the required value by a factor of four for every $201 \leq d \leq 2000$, and the normalized fourth differences for $d \geq 2001$ are at least 6.02 where $1/4$ suffices. A single audit script re-checks all three regimes.

Rigidity

Every step of the proof is an inequality with an explicit defect, so the equality case can be traced back through the chain (Methods). If $I_d = I_{\text{ME}}(d)$, the cone representation forces equality in a continuum inequality for an embedding in which every cell has positive length. This forces equality in the strip inequality at almost

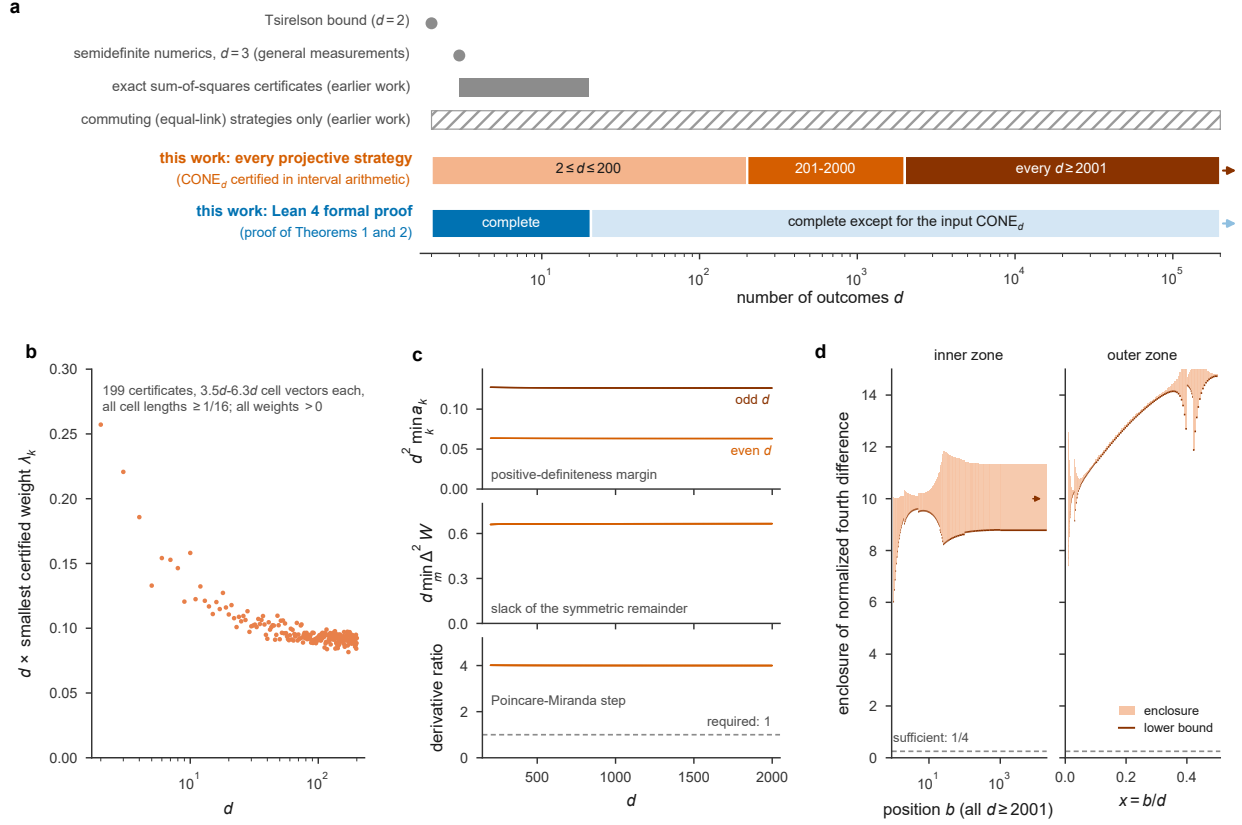


Figure 5. Certification for every d . **a**, Coverage of the optimality statement. Grey: earlier results, namely Tsirelson’s bound ($d = 2$), semidefinite numerics for $d = 3$ with general measurements³⁰, our exact sum-of-squares certificates ($3 \leq d \leq 20$)²⁸ and the commuting (equal-link) case for every d ³¹ (hatched). Orange: the three regimes of the CONE_d certificate, which together with Theorem 3 give Theorem 1 for every d . Blue: the Lean formalisation of the proof of Theorems 1 and 2, complete for $2 \leq d \leq 20$ and, for $d \geq 21$, complete except for the input CONE_d (Theorems 3 and 4 are formalised in full). **b**, $2 \leq d \leq 200$: d times the smallest certified weight of the exact cone certificate (rational cells, interval arithmetic). **c**, $201 \leq d \leq 2000$: certified lower bounds, read from the single-run certificate of each d (160-bit interval arithmetic, every quantity enclosed on one Poincaré–Miranda box per d); plotted are the lower endpoints of the certified enclosures of $d^2 \min_k a_k$, the positive-definiteness margin of the Gaussian modulation (top; even and odd d), of $d \min_m \Delta^2 W$, the slack of the symmetric remainder (middle), and of the derivative bound of the Poincaré–Miranda step divided by its required value (bottom; 1 suffices). The smallest values are 0.06319 (even d), 0.12638 (odd d), 0.660 and 4.0009. **d**, $d \geq 2001$: interval enclosures (bands) and certified lower bounds (lines) of the normalized fourth differences that control positivity, on the boxes of the inner zone (positions $b \geq 1$ with $b/d \leq 0.013$, for all $d \geq 2001$ at once; the last box covers every $b \geq 1000$) and of the outer zone ($0.011 \leq b/d \leq 0.501$). Every lower bound lies far above the sufficient level $1/4$.

every point of the circle, so by the equality clause of Theorem 3 the matrices $B(\theta)$ and $g(\theta)$ commute almost everywhere. Because g is an explicit sum of logarithms of the distances to the cell endpoints, a residue argument then shows that all $4d$ spectral projections commute. In the commuting case our classical theorem³¹ identifies the configuration as a direct sum of DKZ blocks, which gives Theorem 2. Theorem 2 extends to every d the rigidity that our exact certificates established for $3 \leq d \leq 20$ ²⁸; for those d the two proofs are independent.

The status of Open Quantum Problem 27

Part A of Open Quantum Problem 27 asked whether every facet of the local polytope is of CGLMP type; Bancal, Gisin and Pironio answered it negatively⁴⁶. Part B asks to show that the DKZ measurements are necessarily optimal on maximally entangled states, that they give the highest resistance to noise, and that they are optimal for the Kullback–Leibler discrimination of quantum from local correlations¹⁸. The clauses go back to Gill’s numerical observations¹⁷. Table 1 gives the status of each of them. Theorems 1 and 2 prove optimality and uniqueness for every d .

The answer to the noise clause depends on how noise resistance is measured and on which measurements are allowed. Mixing the correlations with uniformly random outcomes of weight $1 - \nu$ multiplies I_d by ν , so the smallest visibility at which the CGLMP inequality detects nonlocality of a maximally entangled state is $2/I_{\text{ME}}(d)$, and by Theorem 2 only the DKZ measurements reach it; this consequence is also checked in Lean. Gill, however, asked for the violation of local realism, that is, of any Bell inequality¹⁷. In this literal reading the clause fails for every $d \geq 4$ once measurements may have outcomes that never occur (Supplementary Information). A strategy that uses the optimal two-outcome CHSH measurements, and never produces the other $d - 2$ outcomes, tolerates more noise than the DKZ measurements: the noise lands partly on the unused outcomes, and this helps the noisy correlations to violate a coarse-grained CHSH inequality. We determine its critical visibility exactly, for example $4(d - 1)/((\sqrt{2} - 1)d^2 + 4(d - 1))$ for even d , and prove that the DKZ measurements need a higher one, with computer-assisted certificates for $4 \leq d \leq 9$ and an elementary bound for $d \geq 10$. The effect itself was noticed before: Acín, Durt, Gisin and Latorre found it for a two-qubit entangled state embedded in two qudits and concluded that resistance to noise is not a good measure of nonlocality²⁴, and Baek, Ryu and Lee observed it numerically on maximally entangled states of local dimension 4 and 16⁴⁷. As far as we found, the general theorem and its bearing on Problem 27 are new. For $d = 2$ the clause holds, and for $d = 3$ it holds on Φ_3 but fails on Φ_2 . The counterexample needs unused outcomes. For complete von Neumann measurements on Φ_d , which use every outcome and for which uniformly random outcomes amount to white noise on the state, the noise clause for all Bell inequalities remains open for $d \geq 4$; numerical searches support it for $d \leq 8$ (ref. 48 and Supplementary Information).

The Kullback–Leibler clause⁴⁹ is false for every $d \geq 4$: an explicit strategy built from pairs of CHSH measurements has a statistical strength of at least 0.0703204 bits, while that of the DKZ measurements is at most 0.0687803 bits in every dimension. Zhang found the first counterexample, for $d = 4$ ⁵⁰, and we gave counterexamples for $4 \leq d \leq 9$ in earlier work²⁸. For $d = 3$ the DKZ measurements are a strict local maximum of the statistical strength; whether they are the global maximum is open. When the state may also be chosen, the best tests use states that are not maximally entangled⁵¹.

Table 1. Status of Open Quantum Problem 27. “Max-ent” refers to maximally entangled states of any local dimension. The positive statements concern projective measurements; for $3 \leq d \leq 8$ optimality and uniqueness also hold for arbitrary POVMs (exact certificates, ref. 58). The counterexample for all Bell inequalities needs measurements with unused outcomes; the Kullback–Leibler counterexamples are complete von Neumann measurements. The results of this work have not yet been reviewed by outside experts.

clause	status
27A: all facets of the local polytope of CGLMP type	false ⁴⁶
27B: DKZ optimal on max-ent states	proved for every d (Theorem 1), with a computer-assisted step for $d \geq 21$; formalised in Lean with no hypotheses for $d \leq 20$; earlier: $3 \leq d \leq 20$ and the commuting case for every d ^{28,31}
27B: DKZ unique (rigidity)	proved for every d (Theorem 2), with the same computer-assisted step for $d \geq 21$; formalised in Lean with no hypotheses for $d \leq 20$; earlier: $3 \leq d \leq 20$ ²⁸
27B: highest noise resistance, CGLMP inequality, uniformly random outcomes (or white noise) as noise	proved for every d , DKZ the unique optimum (Theorems 1 and 2; formalised in Lean with the same scope)
27B: highest noise resistance, all Bell inequalities, complete von Neumann measurements on Φ_d (uniformly random outcomes = white noise on the state)	open for $d \geq 4$; true for $d = 2, 3$; DKZ optimal in numerical searches for $d \leq 8$
27B: highest noise resistance, all Bell inequalities, uniformly random outcomes as noise (Gill’s reading), measurements with unused outcomes allowed	false for every $d \geq 4$ (Supplementary Information); $d = 3$: true on Φ_3 , false on Φ_2
27B: optimal Kullback–Leibler discrimination	false for every $d \geq 4$ (Supplementary Information; $d = 4$ first in ref. 50); $d = 3$: open, DKZ a strict local maximum
maximal value over all states	exact for $3 \leq d \leq 8$ ^{27,28} ; open for $d \geq 9$

Discussion

We have shown that the DKZ measurements are the optimal and essentially unique projective measurements for the CGLMP inequality on maximally entangled states, in every dimension. This answers the clause of IQOQI Open Quantum Problem 27B that asks for necessity on maximally entangled states, which had been supported by numerics for more than two decades. For the noise and Kullback–Leibler clauses our answers are partial (Table 1): the noise clause holds for the CGLMP inequality and is open for all Bell inequalities with complete von Neumann measurements, and the Kullback–Leibler clause fails for $d \geq 4$ and is open for $d = 3$. The proof uses a computer only for the certificate CONE_d , in interval arithmetic, and for $d \leq 20$ it is checked in full by the Lean theorem prover. An analytic proof of CONE_d would remove the computer-assisted step and, once formalised, complete the formal proof for every d . For experiments, Theorem 1 provides the exact benchmark $I_{\text{ME}}(d)$ for maximally entangled sources of every dimension. Theorem 2 shows that projective measurements that reach the benchmark on a maximally entangled source are equivalent to DKZ, the starting point for certifying high-dimensional measurements from the observed statistics. A quantitative version of Theorem 2 would bound the distance to DKZ in terms of the gap $I_{\text{ME}}(d) - I_d$. Our proof suggests a route: it expresses the gap exactly as a positive combination of integrated defects of the strip inequality and of explicit non-negative slack terms, and Theorem 4 expresses each defect as an integral of a positive density that vanishes only in the commuting case. Such a bound is what applications

to randomness certification and robust self-testing would require^{8,34}.

Three questions remain. The first concerns general measurements. Our statements cover projective measurements, which is the setting of the original question. A general measurement on Φ_D can be realized as a projective measurement on a larger space, but the enlarged state is then no longer maximally entangled, so the extension is not automatic. The second concerns the optimal state. The maximal CGLMP value over all states is attained by states that are not maximally entangled²⁴, and it is known exactly only for $d \leq 8$. Our reduction extends to every pure state as a clock model with site-dependent weights (we do not give the details here). Whether the cone method extends to it is open. The third is the noise clause with white noise on the state and every outcome used, which is open for $d \geq 4$.

The mathematical tools may be useful beyond Bell inequalities. The strip inequality is a majorization principle for non-normal matrices whose real part is a projection. It connects the eigenvalues of $B + ig$ with the spectrum of a compression, and it reduces to the Stein–Weiss theorem in the commutative case. Theorem 4 shows that a two-variable, pinched version of the Bessis–Moussa–Villani formula has a positive density that can be written down explicitly, which complements the abstract positivity of Stahl’s theorem^{21,44}. The continuum value Φ_C in equation (10) is the volume of an ideal hyperbolic tetrahedron with dihedral angles $\pi\alpha, \pi\beta, \pi(1 - \alpha - \beta)$ divided by π^2 ⁵², and up to the factor $-1/\pi$ it coincides with the surface tension of lozenge tilings⁵³. We do not know whether these coincidences have a deeper meaning for Bell inequalities.

Methods

CGLMP expression and maximally entangled strategies

We use equation (1), in which $P(A_x = B_y + k)$ means $A_x - B_y \equiv k \pmod{d}$. This is the reading under which local models obey $I_d \leq 2$ and the DKZ measurements give $I_{\text{ME}}(d)$ ¹³. For projective measurements on Φ_D one has $P(a, b|x, y) = \tau(P_a^x(Q_b^y)^\top)$ with $\tau = \frac{1}{D} \text{Tr}$ and the transpose taken in the Schmidt basis of Φ_D . With $w = e^{2\pi i/d}$, $U_x = \sum_a w^a P_a^x$ and $U'_y = \sum_b w^b Q_b^y$, put $R_1 = U_2$, $R_2 = (U'_2)^\top$, $R_3 = U_1$ and $R_4 = (U'_1)^\top$. These are unitaries with $R_i^d = \mathbf{1}$. The Fourier expansion of the residue function $m(t) \in \{0, \dots, d-1\}$ gives

$$I_d = 4 - \frac{2S}{d-1}, \quad S = 2(d-1) + \sum_{n=1}^{d-1} c_n \sum_{i=0}^3 \tau(L_n^{(i)}), \quad c_n = -\frac{1}{1 - w^{-n}}, \quad (11)$$

with the links $L_n^{(0)} = R_1^n R_2^{-n}$, $L_n^{(1)} = R_2^n R_3^{-n}$, $L_n^{(2)} = R_3^n R_4^{-n}$ and $L_n^{(3)} = w^{-n} R_4^n R_1^{-n}$. Conversely, any four unitaries with $R_i^d = \mathbf{1}$ define a projective strategy on Φ_D through their spectral projections³¹.

Reduction to the clock model

Let $z = e^{2\pi i/4d}$. The map $\rho(R_1, R_2, R_3, R_4) = (R_2, R_3, R_4, wR_1)$ leaves S invariant, and since ρ^4 multiplies every R_i by w (which also leaves S invariant), it generates a cyclic group of order $4d$. The direct sum $\tilde{R} = \bigoplus_{r \in \mathbb{Z}_{4d}} \rho^r(R)$ has the same value and contains R as a direct summand. The cyclic shift of the summands is a unitary W with $W\tilde{R}_i W^\dagger = \tilde{R}_{i+1}$ ($i = 1, 2, 3$), $W\tilde{R}_4 W^\dagger = w\tilde{R}_1$ and $W^{4d} = \mathbf{1}$. Then $Z = W^4$ satisfies $Z\tilde{R}_1 Z^\dagger = w\tilde{R}_1$ and $Z^d = \mathbf{1}$. By the Stone–von Neumann theorem for $\mathbb{Z}_d$, $\tilde{R}_1 = X \otimes \mathbf{1}_M$ and $Z = Z_0 \otimes \mathbf{1}_M$, with X the cyclic shift and Z_0 the clock matrix on $\mathbb{C}^d$. Since W commutes with Z , $W = \sum_k |k\rangle\langle k| \otimes z^k V_k$ with $V_k^4 = \mathbf{1}$. Inserting this *covariant* form into equation (11) gives equation (3); conversely every family V

1 defines the covariant strategy $R_1^V = X \otimes \mathbf{1}$, $R_{i+1}^V = WR_i^V W^\dagger$ on $\mathbb{C}^d \otimes \mathbb{C}^M$. The DKZ strategy is, up to the
2 local unitary $W_0^{-2} \otimes \overline{W_0^{-2}}$ with $W_0 = \text{diag}(z^k)$, the covariant strategy with $M = 1$ and $V \equiv 1$. Full proofs are
3 in ref. 31 (Theorem 2.3 and its appendix).

4 The circle configuration and the linear form

5 Let $N = 4d$ and $E_k = V_k^\dagger$. Every $x \in \mathbb{Z}_N$ can be written uniquely as $x = k - da$ with $0 \leq k \leq d - 1$ and
6 $a \in \mathbb{Z}_4$; let Q_x be the spectral projection of E_k for the eigenvalue i^a . For every site k the four projections
7 Q_{k-da} form a projective measurement. Put $A_x = Q_x$, $B_x = Q_{x+d}$, $\kappa(u) = \cot(\pi u/N)$ ($\kappa(0) = 0$), $T(m) =$
8 $\sum_y \tau(A_{y+m} B_y)$ and $N(m) = T(m) - T(-m)$. Then $F(V) = \frac{1}{2} \sum_{x,y} \kappa(x-y) \tau(A_x B_y) - \frac{d}{2}$. Using $N(d) = d$
9 and $N(2d-m) = N(m)$, one finds

$$F(V) - F_{\text{DKZ}} = \frac{1}{2} \langle v, \delta \rangle, \quad v_m = 2 \csc \frac{\pi m}{2d}, \quad \delta_m = N(m) - m \quad (1 \leq m \leq d-1). \quad (12)$$

10 The DKZ configuration is a pair of adjacent blocks and has $\delta = 0$. For $1 \leq s \leq d-1$ an elementary count
11 gives $\delta_s + \delta_{d-s} \leq 0$, so the directions $t_s = e_s + e_{d-s}$ are trivially admissible.

12 Proof of Theorems 3 and 4

13 We write $g_d = BgB + PgP$ and $p_{s,\tau}(x) = \det(g - s - x(P - \tau))$, $q_{s,\tau}(x) = \det(g_d - s - x(P - \tau))$. Complete
14 proofs are in the Supplementary Information; here we list the steps. (i) *Pencil facts*. For $0 < \tau < 1$ every
15 root of $p_{s,\tau}$ satisfies $|x| \leq \|g - s\| / \min(\tau, 1 - \tau)$, and every non-real root satisfies $|x| \leq \|g - s\| / \sqrt{\tau(1 - \tau)}$
16 (a non-real root has an eigenvector ψ with $\langle \psi, (P - \tau)\psi \rangle = 0$). All roots are real if $s \notin (\lambda_{\min}, \lambda_{\max})$, all
17 roots of $q_{s,\tau}$ are real, and $p_{s,\tau}$, $q_{s,\tau}$ have the same leading coefficient and the same sum of roots. Hence F
18 is continuous, bounded by $(M/2\pi)\|g - s\| / \sqrt{\tau(1 - \tau)}$, and compactly supported. (ii) *Jensen-type identity*.
19 If p and q have degree M , the same leading coefficient and the same sum of roots, and q has only real
20 roots, then $\int_{\mathbb{R}} \log |p(\xi)/q(\xi)| d\xi = \pi \sum_i |\text{Im } x_i|$, the sum over the roots of p . This follows from $\int_{\mathbb{R}} \frac{1}{2} \log(1 +$
21 $b^2/u^2) du = \pi|b|$. (iii) *A distribution with Laplace transform $\mathcal{D}/a^2$* . $\mathcal{D}(0, t) = \partial_a \mathcal{D}(0, t) = 0$, so $G = \mathcal{D}/a^2$
22 is entire on $\mathbb{C}^2$, and $|\mathcal{D}(a, t)| \leq 2M \exp \max_{(s,\tau) \in K} (s \text{Re } a - \tau \text{Re } t)$ with $K = [\lambda_{\min}, \lambda_{\max}] \times [0, 1]$. By
23 the Paley–Wiener–Schwartz theorem⁴³ there is a distribution T supported in K with $T(e^{as-t\tau}) = G(a, t)$.
24 (iv) *Fourier slices*. On the line $t = a\xi$, $G(a, a\xi)$ is the Laplace transform of the piecewise linear function
25 $U_\xi(w) = \sum_j [(w - \lambda_j(\xi))_+ - (w - \lambda_j^0(\xi))_+]$, where $\lambda_j(\xi)$ and $\lambda_j^0(\xi)$ are the eigenvalues of $g - \xi P$ and $g_d - \xi P$.
26 The Hilbert transform of U'_ξ is $\frac{1}{\pi} \log |\det(g - \xi P - w)/\det(g_d - \xi P - w)|$, which at $w = s - \xi\tau$ equals
27 $\frac{1}{\pi} \log |p_{s,\tau}(\xi)/q_{s,\tau}(\xi)|$. Fourier inversion in polar coordinates (the Fourier slice theorem⁴²) integrates this
28 over all lines through (s, τ) , and step (ii) gives $T = F$ on the open strip $0 < \tau < 1$. (v) *No mass on*
29 *the edges*. $T - F$ is supported on $\tau \in \{0, 1\}$. By the structure theorem for distributions supported on a
30 line, its Laplace transform is $\sum_k t^k [A_k(a) + e^{-t} B_k(a)]$. The limits $\lim_{t \rightarrow +\infty} \text{Tr } e^{ag-tP} = \text{Tr}_B e^{aBgB}$ and
31 $\lim_{t \rightarrow -\infty} e^t \text{Tr } e^{ag-tP} = \text{Tr}_P e^{aPgP}$ force all A_k and B_k to vanish. Hence $T = F$, which is equation (8).
32 (vi) *Strip inequality*. $h_\lambda(x+iy) = \int (w - \lambda)_+ K_x(y - w) dw$ is the Poisson integral of its boundary values, and
33 $\int e^{-au} K_X(u) du = \sin(a(1 - X))/\sin a$ for $|\text{Re } a| < \pi$. Let ρ be the sum over the eigenvalues $v_k = x_k + iy_k$
34 of $B + ig$ of the harmonic measures $K_{x_k}(y_k - w) dw$, and σ the eigenvalue counting measure of PgP . For
35 $0 < |a| < \pi$, the Laplace transform of ρ is $[\text{Tr } e^{ag+iaP} - \text{Tr } e^{ag-iaP}]/(2i \sin a)$, and that of σ is $\text{Tr}_P e^{aPgP}$.
36 By equation (8) at $t = \mp ia$ the difference is the Laplace transform of $a^2 V$, where $V(w) = \int_0^1 \int K_{1-\tau}(s -$
37 $w) F(s, \tau) ds d\tau \geq 0$. Matching moments then shows that the defect of (5) equals $V(\lambda)$. If the defect vanishes

for one λ , then $F \equiv 0$, hence $\mathcal{D} \equiv 0$, and the second-order term $\mathcal{D}(a, t) = a^2 \|BgP\|_F^2 (1 - e^{-t})/t + O(a^3)$ gives $BgP = 0$, that is, $[B, g] = 0$. (vii) *An elementary route.* Steps (iii)–(v) can be replaced by an argument without distributions, Paley–Wiener theory or Fourier inversion. It rests on a Radon identity: for Hermitian H , with $y_i(\tau)$ the eigenvalues of $(P - \tau)^{-1}H$,

$$\frac{1}{2\pi} \int_0^1 \sum_i |\operatorname{Im} y_i(\tau)| d\tau = \operatorname{Tr} H_+ - \operatorname{Tr}(BHB + PHP)_+. \quad (13)$$

To prove it, shift H to $H + c$ with $\operatorname{Im} c > 0$. Then no root is real and exactly $\operatorname{rank} P$ roots lie in the upper half-plane, and the roots obey the complex inviscid Burgers equation $\partial_\tau y = y \partial_c y$, so that $\partial_\tau \sum \operatorname{Log} y = \partial_c \sum y$ for the roots in the upper half-plane (a Cauchy integral, valid also at repeated roots). Integrating over τ , evaluating the two ends by a Schur complement, and letting c tend to the real axis gives equation (13). Applied to $H = g - \xi P - w$, it says that the integral of F along the line $s = w + \xi\tau$ equals $U_\xi(w)$. Tonelli’s theorem for real a and t and the identity theorem for entire functions then give equation (8) on $\mathbb{C}^2$, and in step (vi) the uniqueness of the Laplace transform is needed only for continuous, exponentially decaying functions of one variable. This is the route followed by the Lean formalisation, which uses the Fourier inversion theorem of Mathlib for that one-variable step. The same argument for an arbitrary Hermitian matrix in place of P gives a proof of Stahl’s theorem²¹ without distribution theory or Fourier analysis; we make no priority claim about this by-product.

The continuum theorem

Let $B(\theta)$ be a projection-valued step function on the circle (step functions are all that the cell embeddings need), $0 \leq A(\theta) \leq \mathbf{1} - B(\theta)$, and g the entrywise conjugate function of B . The Herglotz integral $H(z) = \int \frac{e^{i\phi} + z}{e^{i\phi} - z} B(\phi) \frac{d\phi}{2\pi}$ is analytic on the unit disc with boundary values $B + ig$, and its numerical range, hence its spectrum, lies in the strip. The function $\phi_\lambda(w) = \frac{i}{\pi^2} \operatorname{Li}_2(e^{-i\pi w - \pi\lambda})$ is analytic on the strip with $\operatorname{Re} \phi_\lambda = h_\lambda$, so $z \mapsto \operatorname{Re} \operatorname{Tr} \phi_\lambda(H(z)) = \sum_{\nu \in \operatorname{spec} H(z)} h_\lambda(\nu)$ is harmonic. By the mean-value property and dominated convergence, its boundary mean equals its value at the centre, $\sum_i h_\lambda(\beta_i)$ over the eigenvalues β_i of $\int B$, which is at most $Mh_\lambda(\beta)$ because $x \mapsto h_\lambda(x)$ is concave on $[0, 1]$. With Theorem 3 at every boundary point and $\tau(A(g - \lambda)) \leq \tau((P(g - \lambda)P)_+)$ this gives $\int \tau(Ag) \leq \lambda\alpha + h_\lambda(\beta)$ for every λ . Minimizing over λ gives $\Phi_c(\alpha, \beta)$ by a Legendre identity for the dilogarithm. For $\alpha = \beta = \frac{1}{4}$ the minimizer is $\lambda^* = \log 2/(2\pi)$ and $\Phi_c(\frac{1}{4}, \frac{1}{4}) = G/\pi^2$.

Cell embeddings and CONE_d

A cell vector is $\ell \in \mathbb{R}^d$ with $\ell_r \geq 0$ and $\sum_r \ell_r = d$, extended d -periodically to $\mathbb{Z}_N$. Give the site x the arc $I_x = [L_x, L_{x+1})$ of length ℓ_x on the circle $\mathbb{R}/N\mathbb{Z}$, and let $\tilde{A}, \tilde{B}$ be the step functions equal to A_x, B_x on I_x . Because ℓ depends only on the residue modulo d and every residue class carries a projective measurement, $\int \tilde{B} = \int \tilde{A} = d\mathbf{1}$. The continuum theorem therefore gives $\sum_{x,y} K^\ell(x, y) \tau(A_x B_y) \leq N^2 \Phi_c(\frac{1}{4}, \frac{1}{4})$, with $K^\ell(x, y) = \int_{I_x} \int_{I_y} \cot(\pi(\xi - \eta)/N) d\xi d\eta$, and equality for every DKZ block. Averaging over the d rotations of the configuration gives $\langle u^\ell, \delta \rangle \leq 0$ with

$$u_m^\ell = \Delta_m^2 \frac{1}{d} \sum_{r=0}^{d-1} \hat{G}(\ell_r + \dots + \ell_{r+m-1}), \quad \hat{G}(u) = G(u) + G(2d - u), \quad G(u) = -\frac{N^2}{2\pi^2} \operatorname{Cl}_2\left(\frac{2\pi u}{N}\right), \quad (14)$$

1 where Δ_m^2 is the second difference in m , while $v_m = \hat{G}''(m)$. CONE_d is the statement that $v = \int u^\ell \mu(d\ell) +$
2 $\sum_s w_s t_s$ for a finite positive measure μ carried by cell vectors with all cells positive and numbers $w_s \geq 0$;
3 in particular v lies in the convex cone generated by the vectors u^ℓ and the trivial directions t_s . For cells
4 distributed as d times a Dirichlet(1, ..., 1) vector, the window sums are Beta-distributed, and the Bernstein
5 derivative formula gives $\Delta_m^2 \mathbb{E} \phi(S(m)) = \frac{d}{d+1} \mathbb{E} \phi''(dX)$ with $X \sim \text{Beta}(m+1, d-m+1)$. This reproduces
6 the $1/u$ singularity of $\hat{G}''$ exactly and leaves a smooth residual of relative order $1/d$, which is what the three
7 regimes below correct.

8 **Certification of CONE_d**

9 $2 \leq d \leq 200$. For every d we give $K \approx 3.5d-6.3d$ cell vectors $\ell^{(k)} = n^{(k)}/16$ with integers $n_r^{(k)} \geq 1$ and
10 $\sum_r n_r^{(k)} = 16d$, and weights λ_k with $\sum_k \lambda_k u^{\ell^{(k)}} = v$. All window sums are multiples of $1/16$, so $\hat{G}$ is needed
11 only on a grid. Each Clausen value is enclosed in 160-bit interval arithmetic^{54,55} by its Bernoulli series with
12 an explicit tail bound. The weights come from a floating-point linear program⁵⁶; the exact solution of the
13 linear system is enclosed by a residual bound, and every weight is certified positive (smallest 4.25×10^{-4} ,
14 at $d = 198$; d times the smallest weight is at least 0.0816). A complete re-verification of all 199 certificates
15 passed. Independent floating-point re-evaluations of the identity at selected d , with 50-digit Clausen values
16 and, for small d , the original cell-pair kernel, agree to the precision of the stored weights. For $d \leq 20$ the
17 same certificates are re-evaluated by the Lean kernel in exact rational interval arithmetic.

18 $201 \leq d \leq 2000$. Let η be a centred stationary Gaussian vector on $\mathbb{Z}_d$ whose covariance is the second
19 difference of a variogram Φ , and let $\ell = d \cdot \text{Dirichlet}(1 + \eta)$ on the event $\max_r |\eta_r| < \frac{3}{4}$ (uniform Dirichlet
20 otherwise). The cone identity reduces to two conditions: an antisymmetric part, which decouples up to
21 an error e^{-cd} into one scalar equation per distance $m < d/2$, and a symmetric remainder whose second
22 differences must be non-negative. A Poincaré–Miranda argument on a box around the solution Φ_* of the
23 decoupled equations then produces an exact solution. For each d , a single run of one checker certifies all of
24 this: in 160-bit interval arithmetic throughout, with one code version (its SHA-256 hash is stored in every
25 certificate), it encloses Φ_* (an exact order-14 Gaussian expansion with an order-16 remainder), fixes one
26 Poincaré–Miranda box, and certifies on that widened box that every Fourier coefficient a_k of the variogram
27 is positive (so that it defines a covariance), that the symmetric remainder is positive, that the derivative bound
28 and the sign conditions on the faces of the box hold, and that the Gaussian tail ϵ_t is negligible. Factorials,
29 binomial coefficients and Bernoulli numbers are exact, the zeta values are enclosed by the Euler–Maclaurin
30 formula with a rigorous remainder, and no decision is taken on floating-point numbers. An audit script re-
31 checks all 1800 certificates from the stored exact values (1800 of 1800 pass). The smallest certified margins
32 are $d^2 \min_k a_k \geq 0.06319$ (even d) and ≥ 0.12638 (odd d), $d \min_m \Delta^2 W \geq 0.660$ for the symmetric
33 remainder, a derivative bound 4.0009 times the required value, face margins above $10^{16.7}$ times the required
34 $2\epsilon_t$, and $\epsilon_t \leq 10^{-31.16}$.

35 $d \geq 2001$. The same process is analyzed for all $d \geq 2001$ at once, with $\epsilon = 1/d$. A smooth model
36 Φ_e is defined through the root of a quartic equation with explicit special-function coefficients. Its fourth
37 differences, which control the positive definiteness through a Fejér–Pólya argument, are enclosed by interval
38 Taylor models in (x, ϵ) on boxes covering $0.011 \leq x \leq 0.501$ (outer zone, lower bound 7.419) and all
39 stencil positions with $x \leq 0.013$ (inner zone, lower bound 6.022). Separate bounds hold at the first two
40 stencil positions. The decoupled solution differs from the model by at most $6.6468 \times 10^{-4} \epsilon^2$ pointwise,
41 which yields $a_k \geq 0.1186944 w_k \epsilon^2$ for all k and $\Delta^2 W \geq 0.620154 \epsilon$ for the symmetric remainder; the tail

and fixed-point conditions hold with logarithmic margins above 800. The constants entering these bounds, such as the Fejér–Pólya constant 0.1054878997577, are themselves computed in interval arithmetic. A final rigour pass completed the written proofs of the model identities and corrected some code-level enclosure defects, one of which had left a sliver of size about 10^{-20} of the parameter range uncovered at $d = 2001$; no conclusion changed. One command re-checks all three regimes (the coverage of every box and every margin) and reports CONE_d certified for every $d \geq 2$.

Rigidity

Let $I_d = I_{\text{ME}}(d)$, so $\langle v, \delta \rangle = 0$. In the cone representation every term is non-positive, so $\langle u^\ell, \delta \rangle = 0$ for a cell vector ℓ with all cells positive. Every certificate contains such vectors: for $d \leq 200$ all cells are at least $1/16$, and for $d > 200$ the Dirichlet law charges only the open simplex. Then every rotated continuum inequality is tight. Following the continuum proof, equality forces equality in Theorem 3 at λ^* for almost every θ , hence $[B(\theta), g(\theta)] = 0$ almost everywhere. On each open arc I_x , $g(\theta) = \frac{1}{\pi} \sum_j (B_j - B_{j-1}) \log |\sin \frac{\theta - t_j}{2}|$ is real-analytic, with distinct endpoints t_j . Differentiating $[B_x, g(\theta)] = 0$ and comparing the residues at the poles t_j shows that $[B_x, B_j]$ does not depend on j ; it therefore vanishes, so all Q_x commute. Then the V_k commute, the strategy has equal links, and Theorem B and Corollary C of ref. 31 identify it as $\text{DKZ} \otimes \mathbf{1}$ up to $u \otimes \bar{u}$.

Re-verification

Four separate re-verifications were carried out. In each, the proof was re-derived step by step, including all signs and conventions, and the numerical checks were rewritten from scratch; the reports are in the repository. These are internal checks, not peer review. (1) Theorems 3 and 4 by the Fourier-slice route: correct. The re-implementation confirms equation (8) at 82 complex points (a, t) to 6.8×10^{-13} and the defect formula at 51 pairs of instance and λ to 1.3×10^{-12} , including adversarial configurations with spectra up to ± 150 and clustered eigenvalues. (2) The elementary proof of the Radon identity (step (vii) above): correct; the identities were confirmed to 34–46 digits. (3) The reduction chain, the continuum theorem, the cell inequalities, the logic of CONE_d and the rigidity argument: correct. (4) A second proof of Theorem 3 for rank $B \leq 1$, which also covers corank $B \leq 3$ and hence all $M \leq 5$, by a one-dimensional reduction and the argument principle: correct (no mathematical error; some expository gaps, none of which changes a statement). In addition, for $d \leq 200$ the CONE_d certificates are checked by a script separate from the one that produced them, in all three regimes the rigorous computations agree with independent floating-point computations, and the rigidity chain was tested numerically on step fields and random non-commuting configurations. For $3 \leq d \leq 20$, Theorems 1 and 2 were proved earlier by exact sum-of-squares certificates that use neither Theorem 3 nor CONE_d ²⁸. The results on the other clauses were checked in the same way: the theorem on the literal noise clause by two re-verifications, and the Kullback–Leibler theorem by one; no error was found.

Formal verification

The argument is formalised in the Lean 4 theorem prover (version 4.33.1) with its mathematical library Mathlib (revision 0df444a)^{22,23}, in 98 files of about 33,400 lines. No file uses `sorry`, `admit`, an `axiom` declaration or `native_decide`, and every main theorem depends only on the three standard axioms of

Lean (propext, Classical.choice, Quot.sound). The main formal theorems state the following. For every $2 \leq d \leq 20$, Theorems 1 and 2 and the value of the DKZ strategy hold, with no assumption (OQP27.maxEntClause_le_twenty). For every $d \geq 2$, Theorems 1 and 2 hold, assuming only CONE_d for $d \geq 21$ (OQP27.maxEntClause_all). Theorem 3 with its equality case, Theorem 4 with the defect formula, and the Radon identity (13) hold for every matrix size, with no assumption; the continuity of F , the sum rule of Fig. 4d and the dilogarithm formula (4), which the formal proof does not need, are proved on paper only. The formal rigidity statement asserts that d divides D and that a unitary u maps the measurements to $\text{DKZ} \otimes \mathbf{1}$; the uniqueness of u and the case $d \nmid D$ follow by short arguments on paper. The remaining formal theorems, all without assumptions, are the exact reduction from strategies to the linear form (12), the continuum theorem for step fields (with explicit Herglotz functions), which gives every cell inequality, the classical (commuting) theorem for every d , the DKZ value for every d , and CONE_d for $2 \leq d \leq 20$, whose explicit certificates are re-evaluated by the Lean kernel in exact rational interval arithmetic by a checker that is itself proved sound. Hence Theorems 1 and 2, in the form stated in Lean, are machine-checked with no hypotheses for $2 \leq d \leq 20$, and for $d \geq 21$ the formal proof is complete except for the single input CONE_d , which is established by the interval-arithmetic certificates described above, outside Lean. In Lean, CONE_d is stated as a finite positive combination of cell vectors that includes a cell vector with all cells positive; for $d \geq 201$ this finite form follows from the integral form produced by the certificates by Carathéodory's theorem, the trivial directions being realized by cells supported on two residues. The consequence for the CGLMP noise threshold (Table 1) is formalised in a further file, CglmpNoise.lean, with the same assumptions as Theorems 1 and 2 and the same three axioms. These formal statements use the internal model of the proof (projective measurements on Φ_D of any local dimension, probabilities $\text{Tr}(A^\top B)/D$, the expression I_d). A second, single-file formalisation (directory formal-conjectures-27/ of the repository) states Part B from scratch in the formulation of Formal Conjectures issue #3444 (the state Φ_d , complete von Neumann measurements, the Born rule and the CGLMP functional of the problem page), checks these definitions on special cases, proves that they reduce to the internal model, and builds inside that repository with its continuous-integration settings.

The noise and Kullback–Leibler clauses

Let $u(a, b|x, y) = d^{-2}$ be the behaviour of uniformly random outcomes, and let $v_c(p)$ be the largest v for which $vp + (1 - v)u$ is local. For even d the competitor measures the optimal CHSH observables on the factor $\mathbb{C}^2$ of $\Phi_d = \Phi_2 \otimes \Phi_{d/2}$ and records their two outcomes as 0 and 1; for odd d a block construction adds one dimension with the fixed outcome 0. Coarse-graining every measurement into the outcome 0 and the rest gives a CHSH inequality that is violated exactly for $v > 4(d - 1)/((\sqrt{2} - 1)d^2 + 4(d - 1))$ (even d), respectively $v > 4/((\sqrt{2} - 1)d + 4)$ (odd d), and an explicit local model shows that the noisy behaviour is local at the threshold. For the DKZ measurements, every behaviour with uniform marginals is local at $v = \frac{1}{2}$, which settles $d \geq 10$; for $4 \leq d \leq 9$, exact rational local models, verified with rigorous enclosures of the trigonometric probabilities, give $v_c \geq 0.69, 0.687, 0.684, 0.683, 0.682, 0.681$, above the values of the competitor. The thresholds $2/I_{\text{ME}}(d)$ of the DKZ measurements were known numerically^{12,13,16,48}, and analytically for $d = 3$ ⁵⁷. For the Kullback–Leibler clause the competitor combines copies of the CHSH strategy, and of the DKZ strategy for $d = 3$, on blocks with disjoint outcomes, and the bound for the DKZ measurements uses that their correlations for every d are the reduction modulo d of one probability distribution on the integers. Statements, proofs, certificates and the re-verification reports are in the Supplementary

Scope

All statements concern projective measurements on maximally entangled states Φ_D of finite dimension D . Positive operator-valued measurements are not covered. The clock-model reduction extends to all pure states (we do not give the details here), but the cone method has not been applied there. The counterexamples to the noise and Kullback–Leibler clauses use projective measurements on Φ_d ; the one for the noise clause has unused outcomes, and a variant with positive operator-valued measurements in which every outcome occurs also beats DKZ (Supplementary Information). For complete von Neumann measurements, the noise clause for all Bell inequalities is open for $d \geq 4$.

Data availability

The certificate files for CONE_d in all three regimes, the audit and verification logs, and all data shown in the figures are available at <https://github.com/anshM123/IQOQI-OQP-27> (ref. 58). The certificates for the noise and Kullback–Leibler clauses and the clause-by-clause ledger of Problem 27 are in its directory `clauses/`. The exact sum-of-squares certificates of our earlier work for $3 \leq d \leq 20$ are available at <https://github.com/anshM123/CGLMP> (ref. 28).

Code availability

The certificate generators and checkers for CONE_d , the audit scripts, the Lean 4 formalisations, the re-verification reports with their numerical checks, and the script that produces Figs. 1–5 are available at <https://github.com/anshM123/IQOQI-OQP-27> (ref. 58). The verification programs for the noise and Kullback–Leibler clauses are in its directory `clauses/` (subdirectories `noise-literal/` and `kl-divergence/`), and the Lean file for the CGLMP noise threshold is `lean/OQP27/CglmpNoise.lean`.

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Author contributions 28

A.M. conceived and carried out the project, developed the proofs and the certificates, performed the numerical checks and wrote the manuscript. A.S. validated the mathematics. 29

Competing interests 31

The authors declare no competing interests. 32

Additional information 33

Supplementary Information is available for this paper. It contains complete proofs of Theorems 3 and 4 and an outline of their elementary proof, the precise statements of the reduction, the continuum theorem, the cell inequalities and the rigidity argument, the inventory of certificates, the theorems on the noise and Kullback–Leibler clauses with their proofs, the list of the main Lean theorems, and the re-verifications and checks. **Correspondence and requests for materials** should be addressed to Ansh Mishra (ansh.mishra2025@gmail.com). 34